

CPE333 PROBLEM SESSION 2: Process Creation & Pipes.

1. Fork

1.1 Hello World Program 1.c

Code

```
#include <stdio.h>
#include <unistd.h>

int main(void)
{
    fork();

    printf("Hello World! PID = %d\n", getpid());

    return 0;
}
```

Results

```
Hello World! PID = 11241
Hello World! PID = 11242
```

Discussion

The `fork()` is called, so two processes are created and we get two "Hello World" outputs. The first process is the parent and the second is the child. The parent already exists before, so it was assigned PID 11241, while the child was assigned PID 11242.

1.2 Fork with wait() 2.c

Code

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
    fork();

    wait(NULL);

    printf("Hello World! PID = %d\n", getpid());

    return 0;
}
```

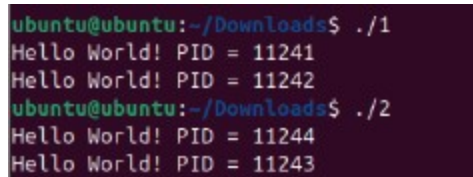
Results

```
Hello World! PID = 11244
Hello World! PID = 11243
```

Discussion

After adding `wait()` , the parent waits for the child to finish. Therefore, the child prints its PID first, and the parent prints its PID afterward. Making the first PID 11244 from the child and 11243 from the parent.

Output Screenshot



```
ubuntu@ubuntu:~/Downloads$ ./1
Hello World! PID = 11241
Hello World! PID = 11242
ubuntu@ubuntu:~/Downloads$ ./2
Hello World! PID = 11244
Hello World! PID = 11243
```

2. Fork Scenarios

2.1 Parent Dies First 3.c

Code

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>

// parent die first
int main(void)
{
    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child: PID = %d, initial PPID = %d\n", getpid(), getppid());

        sleep(4);

        printf("Child: New PPID = %d\n", getppid());
        printf("Child not dead yet...\n");
        system("ps -f");
    }
    else
    {
        printf("Parent: PID = %d\n", getpid());
        printf("Parent not dead yet...\n");
        sleep(2);
        printf("The parent was brutally murdered, but the child is still running.\n");
        exit(0);
    }
    return 0;
}
```

Results

Child: PID = 9761, initial PPID = 9760

Parent: PID = 9760

Parent not dead yet...

The parent was brutally murdered, but the child is still running.

Child: New PPID = 2212

Child not dead yet...

UID	PID	PPID	C	STIME	TTY	TIME	CMD
ubuntu	5685	5672	0	14:33	pts/0	00:00:01	bash
ubuntu	9761	2212	0	16:17	pts/0	00:00:00	./3
ubuntu	9762	9761	0	16:17	pts/0	00:00:00	sh -c -- ps -f
ubuntu	9763	9762	99	16:17	pts/0	00:00:00	ps -f

Discussion

The parent has PID 9760, while the child has PID 9761 and initially has a PPID of 9760, showing that 9760 is its parent. After the parent calls `exit()`, the child continues running even though its parent has terminated. Its PPID then changes from 9760 to 2212, indicating that the child has been adopted by another process.

2.2 Child Dies First 4.c

Code

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

// child die first
int main(void)
{
    pid_t pid = fork();
    wait(NULL);
    if (pid == 0)
    {
        printf("Child: PID = %d\n", getpid());
        printf("Child not dead YET...\n");

        printf("The child fell from high place, but the parent did nothing and is still running\n");
    }
    else
    {
        printf("Parent: PID = %d, initial PPID = %d\n",getpid(), getppid());
        printf("Parent not dead yet... and is waiting for the child to die.\n");
        printf("Parent: New PPID = %d\n", getppid());
        printf("Parent is left alive alone...\n");
        system("ps -f");
    }
    return 0;
}
```

Results

Child: PID = 9765

Child not dead YET...

The child fell from high place, but the parent did nothing and is still running outside.

Parent: PID = 9764, initial PPID = 5685

Parent not dead yet... and is waiting for the child to die.

Parent: New PPID = 5685

Parent is left alive alone...

UID	PID	PPID	C	STIME	TTY	TIME	CMD
ubuntu	5685	5672	0	14:33	pts/0	00:00:01	bash
ubuntu	9764	5685	0	16:17	pts/0	00:00:00	./4
ubuntu	9766	9764	0	16:17	pts/0	00:00:00	sh -c -- ps -f
ubuntu	9767	9766	0	16:17	pts/0	00:00:00	ps -f

Discussion

The parent has PID 9764, while the child has PID 9765. The child terminates first, while the parent continues running. The parent stays alive, and its PPID remains 5685 because the shell is still running. The `wait()` call causes the parent to wait until the child terminates before continuing.

Output Screenshot

```
ubuntu@ubuntu:~/Downloads$ ./3
Child: PID = 9761, initial PPID = 9760
Parent: PID = 9760
Parent not dead yet...
The parent was brutally murdered, but the child is still running.
ubuntu@ubuntu:~/Downloads$ Child: New PPID = 2212
Child not dead yet...


| UID    | PID  | PPID | C  | STIME | TTY   | TIME     | CMD            |
|--------|------|------|----|-------|-------|----------|----------------|
| ubuntu | 5685 | 5672 | 0  | 14:33 | pts/0 | 00:00:01 | bash           |
| ubuntu | 9761 | 2212 | 0  | 16:17 | pts/0 | 00:00:00 | ./3            |
| ubuntu | 9762 | 9761 | 0  | 16:17 | pts/0 | 00:00:00 | sh -c -- ps -f |
| ubuntu | 9763 | 9762 | 99 | 16:17 | pts/0 | 00:00:00 | ps -f          |

ubuntu@ubuntu:~/Downloads$ ./4
Child: PID = 9765
Child not dead YET...
The child fell from high place, but the parent did nothing and is still running outside.
Parent: PID = 9764, initial PPID = 5685
Parent not dead yet... and is waiting for the child to die.
Parent: New PPID = 5685
Parent is left alive alone...


| UID    | PID  | PPID | C | STIME | TTY   | TIME     | CMD            |
|--------|------|------|---|-------|-------|----------|----------------|
| ubuntu | 5685 | 5672 | 0 | 14:33 | pts/0 | 00:00:01 | bash           |
| ubuntu | 9764 | 5685 | 0 | 16:17 | pts/0 | 00:00:00 | ./4            |
| ubuntu | 9766 | 9764 | 0 | 16:17 | pts/0 | 00:00:00 | sh -c -- ps -f |
| ubuntu | 9767 | 9766 | 0 | 16:17 | pts/0 | 00:00:00 | ps -f          |


```

3. Fork Counting 5.c

3.1 Singular Program

Code

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
    int count = 0;

    while (1)
    {
        pid_t pid = fork();
        count++;
        printf("Successfully called fork() %d times.\n", count);
        wait(NULL);
    }

    return 0;
}
```

Results

Successfully called fork() 2874 times.

Output Screenshot:

```
testSuccessfully called fork() 2863 times.  
testSuccessfully called fork() 2863 times.  
testSuccessfully called fork() 2864 times.  
testSuccessfully called fork() 2864 times.  
testSuccessfully called fork() 2865 times.  
testSuccessfully called fork() 2865 times.  
testSuccessfully called fork() 2866 times.  
testSuccessfully called fork() 2866 times.  
testSuccessfully called fork() 2867 times.  
testSuccessfully called fork() 2867 times.  
testSuccessfully called fork() 2868 times.  
testSuccessfully called fork() 2868 times.  
testSuccessfully called fork() 2869 times.  
testSuccessfully called fork() 2869 times.  
testSuccessfully called fork() 2870 times.  
testSuccessfully called fork() 2870 times.  
testSuccessfully called fork() 2871 times.  
testSuccessfully called fork() 2871 times.  
testSuccessfully called fork() 2872 times.  
testSuccessfully called fork() 2872 times.  
testSuccessfully called fork() 2873 times.  
testSuccessfully called fork() 2873 times.  
testSuccessfully called fork() 2874 times.  
testSuccessfully called fork() 2874 times.█
```

3.1 Multiple Program

Results

Successfully called fork() 1730 times.

Discussion

The program was able to successfully call fork() 2,874 times when running alone. When multiple copies of the program were running at the same time, it was able to call fork() 1,730 times. The number was lower because multiple programs were using the system's process resources at the same time.

Output Screenshot:

```
Successfully called fork() 1708 times.  
Successfully called fork() 1709 times.  
Successfully called fork() 1710 times.  
Successfully called fork() 1711 times.  
Successfully called fork() 1712 times.  
Successfully called fork() 1713 times.  
Successfully called fork() 1714 times.  
Successfully called fork() 1715 times.  
Successfully called fork() 1716 times.  
Successfully called fork() 1717 times.  
Successfully called fork() 1718 times.  
Successfully called fork() 1719 times.  
Successfully called fork() 1720 times.  
Successfully called fork() 1721 times.  
Successfully called fork() 1722 times.  
Successfully called fork() 1723 times.  
Successfully called fork() 1724 times.  
Successfully called fork() 1725 times.  
Successfully called fork() 1726 times.  
Successfully called fork() 1727 times.  
Successfully called fork() 1728 times.  
Successfully called fork() 1729 times.  
Successfully called fork() 1730 times.
```

4. Pipe 7.c

Code

```
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
#include <string.h>  
  
// 4 pipe  
int main(void)  
{  
    int fd[2];  
    pipe(fd);
```

```

pid_t pid = fork();
if (pid == 0)
{
    close(fd[0]);

    printf("Child: Child PID: %d\n", getpid());

    char message[30];

    snprintf(message, sizeof(message),
              "Hello from child PID: %d", getpid());

    write(fd[1], message, strlen(message) + 1);

    close(fd[1]);
}
else
{
    close(fd[1]);

    printf("Parent: Parent PID: %d\n", getpid());

    char buffer[99];

    ssize_t bytes_read = read(fd[0], buffer, sizeof(buffer));

    if (bytes_read > 0)
    {
        printf("Parent: %s\n", buffer);
    }

    close(fd[0]);

    wait(NULL);
}

return 0;
}

```

Results

Parent: Parent PID: 10372

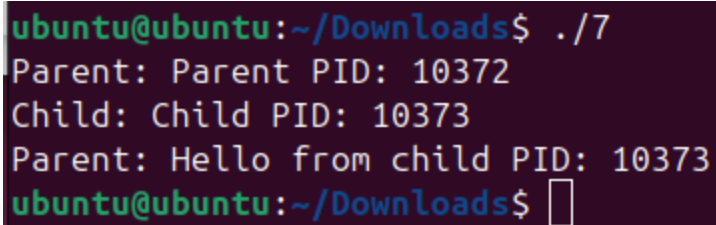
Child: Child PID: 10373

Parent: Hello from child PID: 10373

Discussion

The child `write()` a message containing its PID into the pipe, while the parent `read()` and displays the message. The parent `read()` and displays the child's PID, which is 10373.

Output Screenshot

A terminal window with a dark background and light-colored text. The prompt is 'ubuntu@ubuntu:~/Downloads\$'. The output of the program is displayed in four lines: 'Parent: Parent PID: 10372', 'Child: Child PID: 10373', 'Parent: Hello from child PID: 10373', and the prompt 'ubuntu@ubuntu:~/Downloads\$' followed by a cursor. The text is color-coded: 'Parent:' is green, 'Child:' is red, and the rest is white.

```
ubuntu@ubuntu:~/Downloads$ ./7
Parent: Parent PID: 10372
Child: Child PID: 10373
Parent: Hello from child PID: 10373
ubuntu@ubuntu:~/Downloads$
```

5. Pipe Scenarios

5.1 Sender Reads Its Own Message 8.c

Code

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

// 5.1. sender tries to read back its own message
int main(void)
{
    int fd[2];
    pipe(fd);

    pid_t pid = fork();

    if (pid == 0)
    {
        char buffer[99];

        write(fd[1], "Hello from child", strlen("Hello from child") + 1);

        read(fd[0], buffer, sizeof(buffer));

        printf("Child read: %s\n", buffer);

        close(fd[0]);
        close(fd[1]);
        exit(0);
    }
    else
    {
        close(fd[0]);
        close(fd[1]);
        wait(NULL);
    }
}
```

```
    return 0;  
}
```

Results

```
Child read: Hello from child
```

Discussion

The child `write()` a message into the pipe and then `read()` from the same pipe. Since the message is stored in the pipe, the child is able to `read()` back its own message, displaying "Hello from child".

5.2 Receiver Reads Before Sender 9.c

Code

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

//5.2. receiver reads before sender sends
int main(void)
{
    int fd[2];
    pipe(fd);

    pid_t pid = fork();

    if (pid == 0)
    {
        close(fd[0]);

        sleep(2);

        write(fd[1], "Hello from child", strlen("Hello from child") + 1);

        close(fd[1]);
        exit(0);
    }
    else
    {
        char buffer[99];

        close(fd[1]);

        printf("Parent trying to read...\n");

        read(fd[0], buffer, sizeof(buffer));

        printf("Parent received: %s\n", buffer);

        close(fd[0]);
        wait(NULL);
    }
}
```

```
}  
  
    return 0;  
}
```

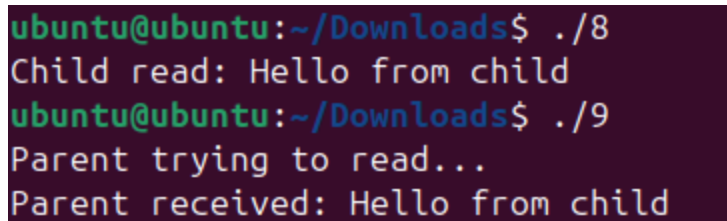
Results

```
Parent trying to read...  
Parent received: Hello from child
```

Discussion

The parent tries to `read()` from the pipe before the child sends its message. Since the pipe is empty, the `read()` function waits until the child writes data into the pipe. After 2 seconds, the child sends the message, and the parent receives and displays it.

Output Screenshot



```
ubuntu@ubuntu:~/Downloads$ ./8  
Child read: Hello from child  
ubuntu@ubuntu:~/Downloads$ ./9  
Parent trying to read...  
Parent received: Hello from child
```


5.3 Sender Sends Multiple Messages Before Receiver Reads

(10.c)

Code

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main(void)
{
    int fd[2];
    pipe(fd);

    pid_t pid = fork();

    if (pid == 0)
    {
        close(fd[0]);

        write(fd[1], "Hello from child 1", strlen("Hello from child 1") + 1);
        write(fd[1], "Hello from child 2", strlen("Hello from child 2") + 1);
        write(fd[1], "Hello from child 3", strlen("Hello from child 3") + 1);

        close(fd[1]);
        exit(0);
    }
    else
    {
        char buffer[99];

        close(fd[1]);

        sleep(2);

        int n = read(fd[0], buffer, sizeof(buffer) - 1);
        buffer[n] = '\0';

        printf("Parent received:\n%s", buffer);

        close(fd[0]);
        wait(NULL);
    }
}
```

```
}

return 0;
}
```

Results

The code above

```
Parent received:
Hello from child 1
```

Remove the + 1 after teh strlen()

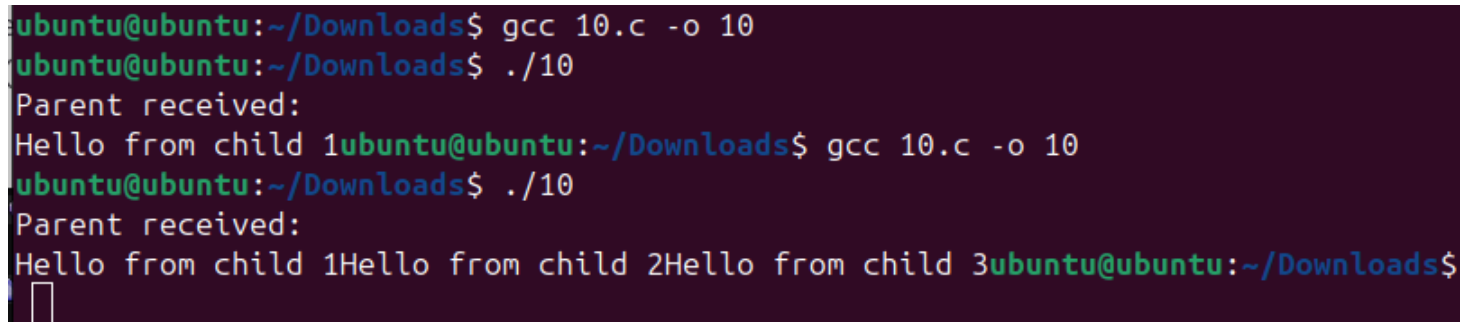
```
Parent received:
Hello from child 1Hello from child 2Hello from child 3
```

Discussion

The child sends three separate messages into the pipe before the parent `read()` them. The messages are stored in the pipe until the parent calls `read()`. With + 1 included in each `write()` , each message contains a null terminator (`\0`), so the parent only displays the first message.

When + 1 is removed, no null terminator is included between the messages, so the parent displays all three messages together.

Output Screenshot



```
ubuntu@ubuntu:~/Downloads$ gcc 10.c -o 10
ubuntu@ubuntu:~/Downloads$ ./10
Parent received:
Hello from child 1ubuntu@ubuntu:~/Downloads$ gcc 10.c -o 10
ubuntu@ubuntu:~/Downloads$ ./10
Parent received:
Hello from child 1Hello from child 2Hello from child 3ubuntu@ubuntu:~/Downloads$
```